Brac University
Department of Electrical & Electronic Engineering
Spring2026

Course Number : EEE308L

Section :06

Group No :04

PSpice Lab Report 2

Name: Souvik Barman Ratul

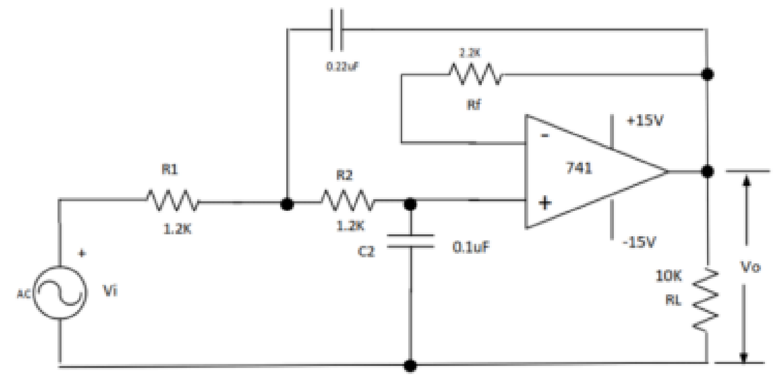
ID:24121205

Section: 06

Semester:-Spring 2026

Problem-1

- 40 dB/Decade Low Pass Butterworth Filter



Draw the circuit in and give the source 5V amplitude an 0 phase.

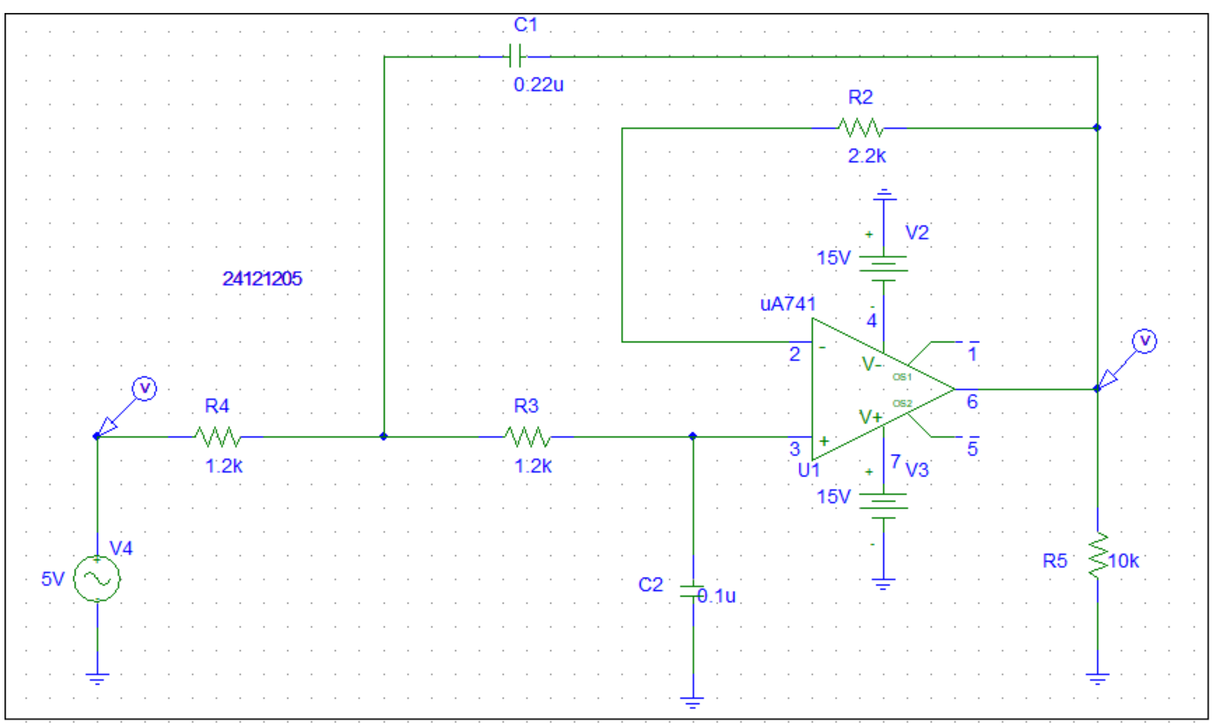
(i) Plot the Gain and Gain(dB) with respect to frequency. Use a suitable frequency range.

(ii) From both the Gain and Gain(dB) plot, find the cutoff frequency.

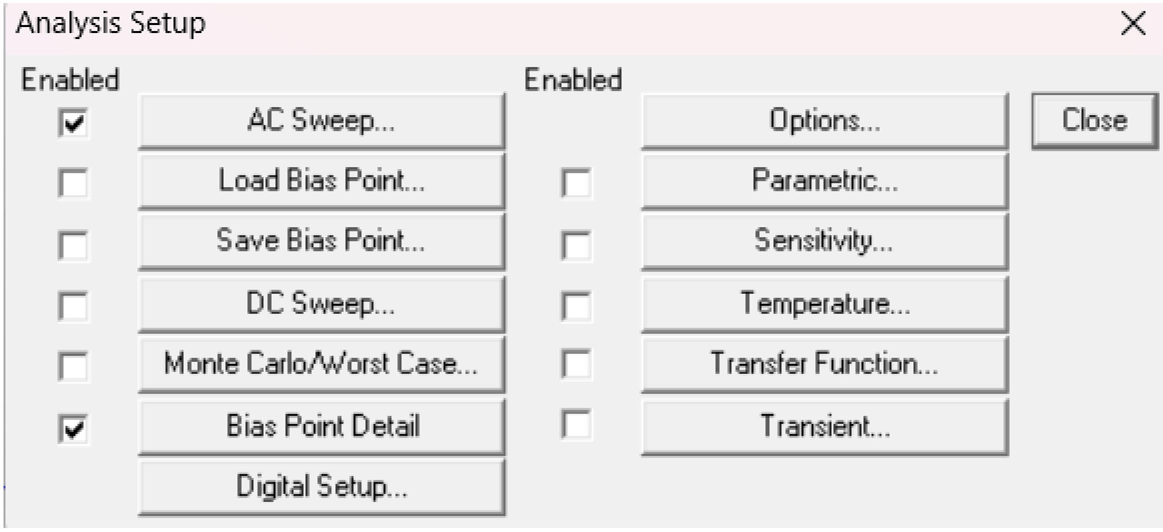
(iii) Show that the roll-off is -40dB/Decade.

Answer:-

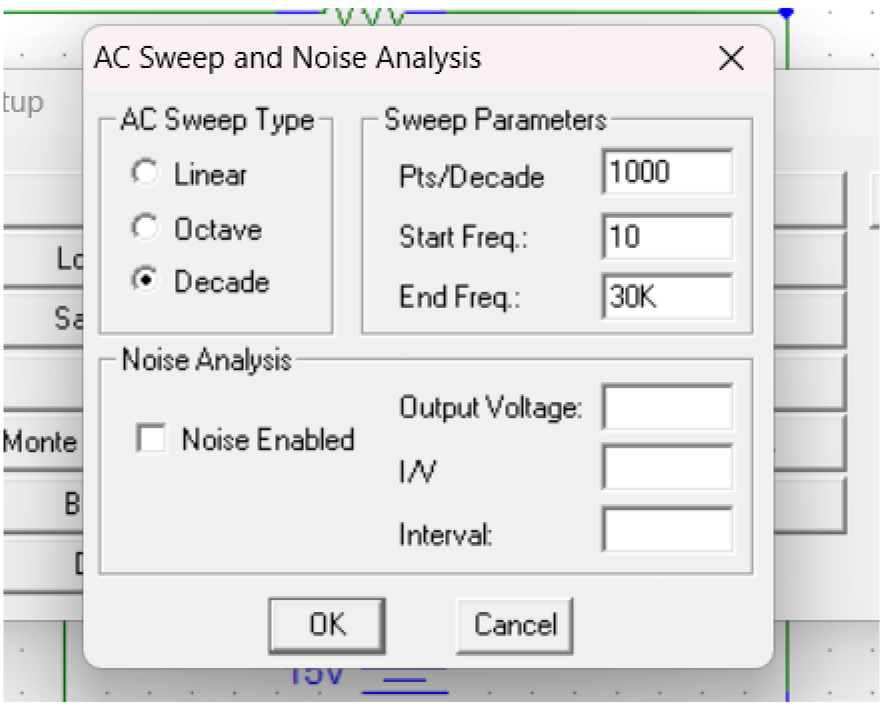
(i) **Circuit Diagram-**



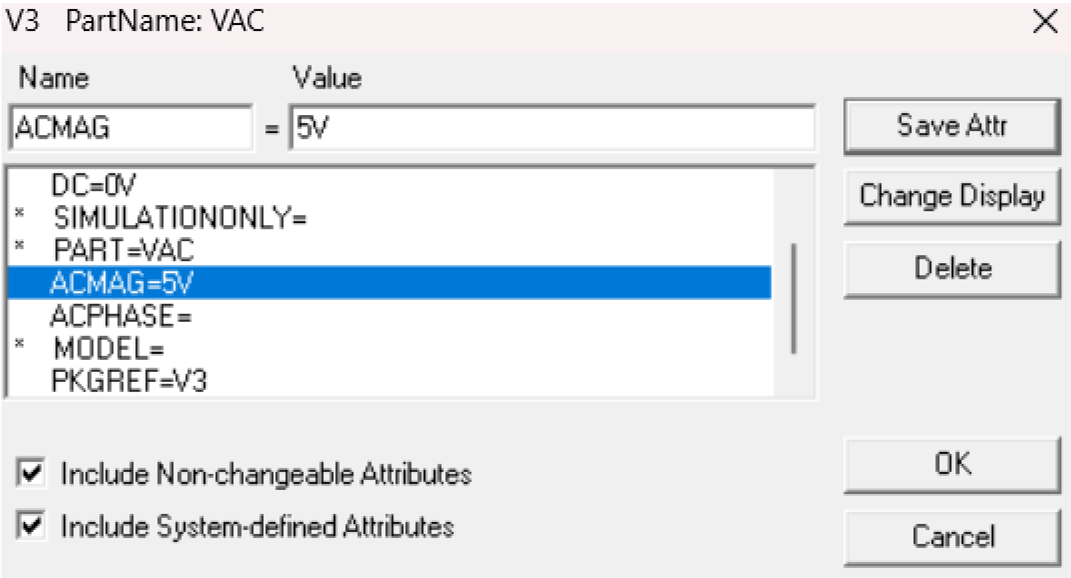
Analysis Setup:



AC Sweep Setup



VAC parameter



Graph

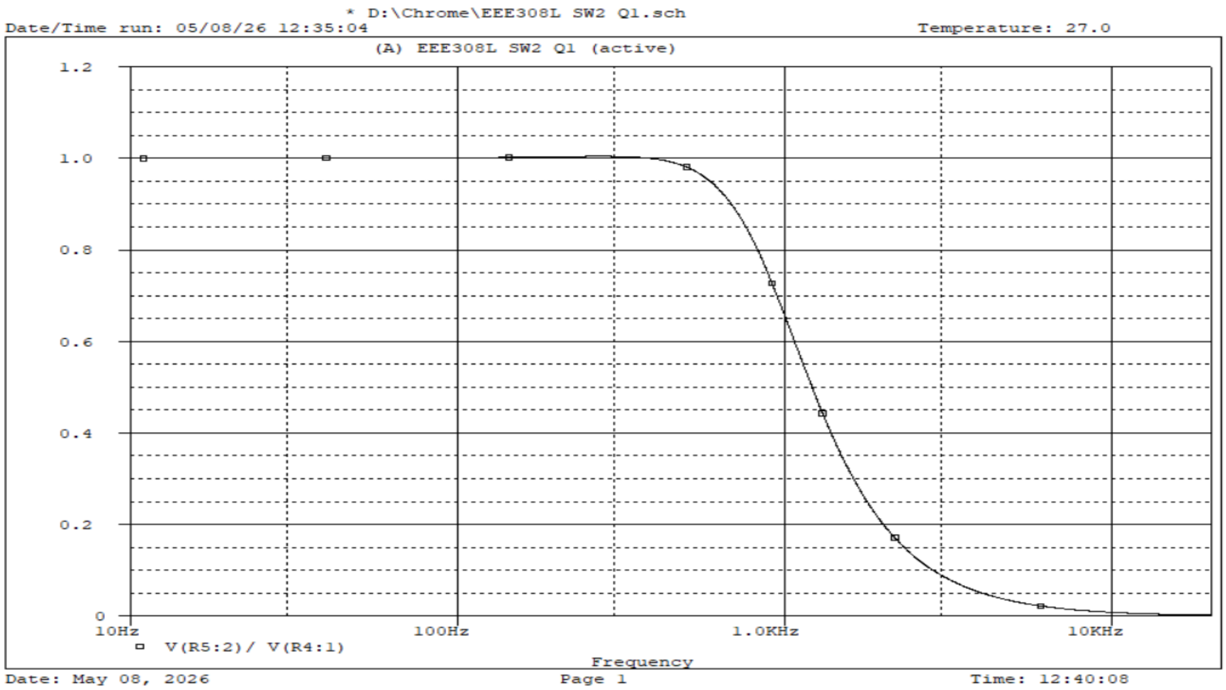


Figure: Graph for Gain

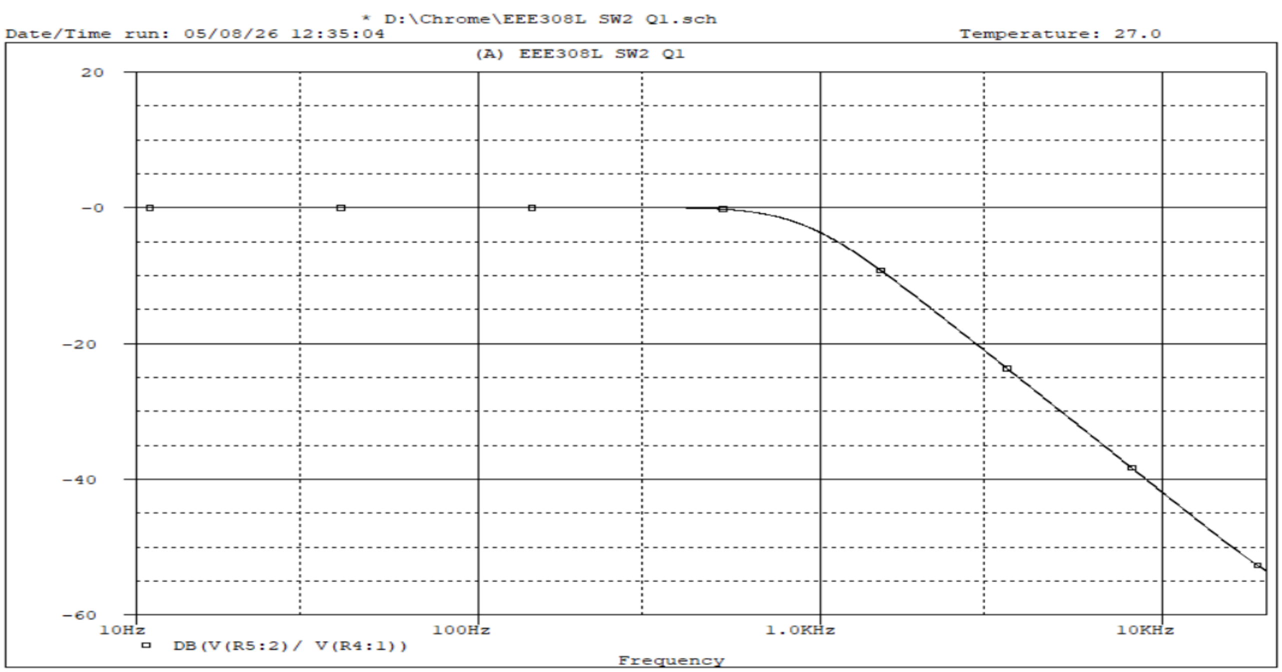


Figure: Graph for Gain (dB)

(ii)

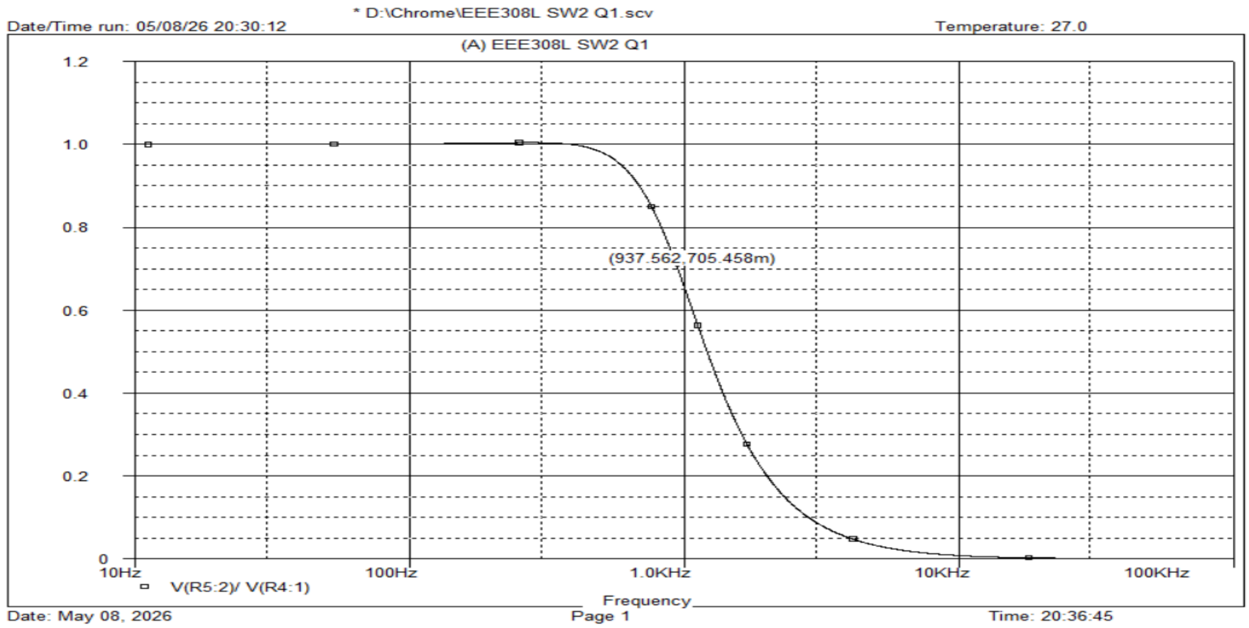


Figure: Graph for Gain (cutoff f= 937.562Hz)

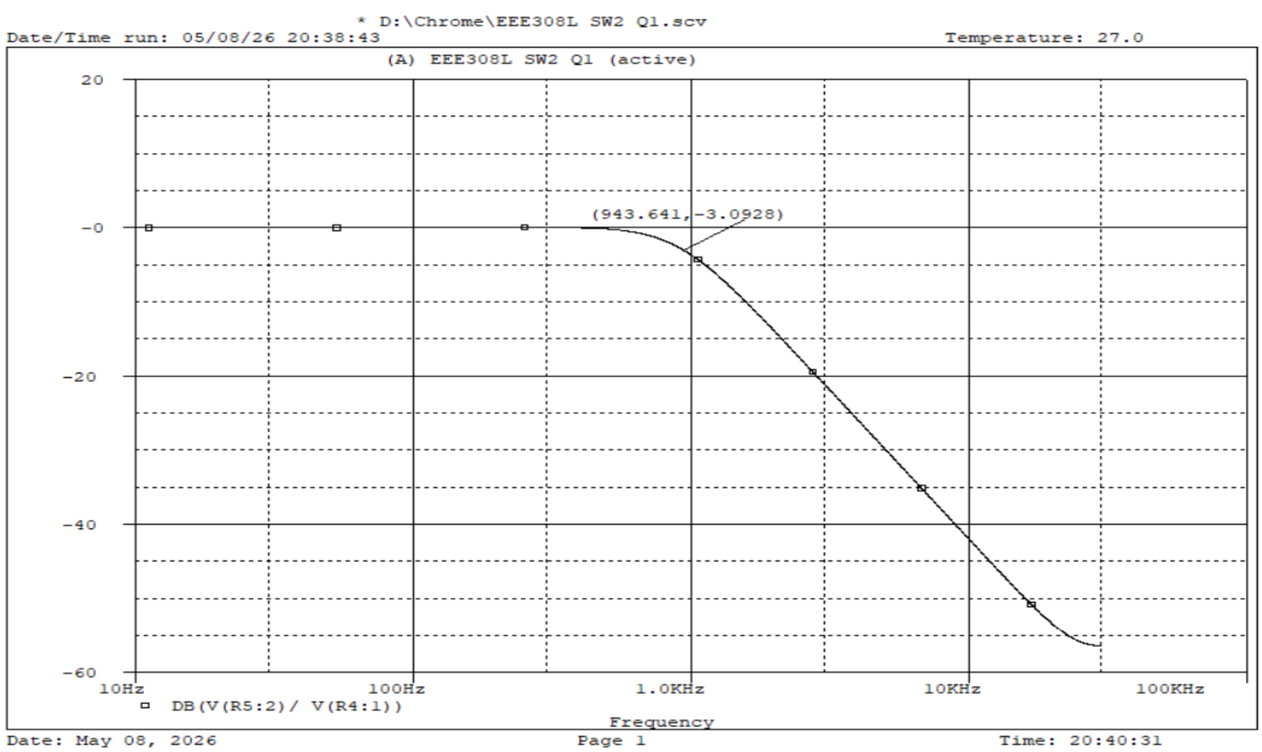
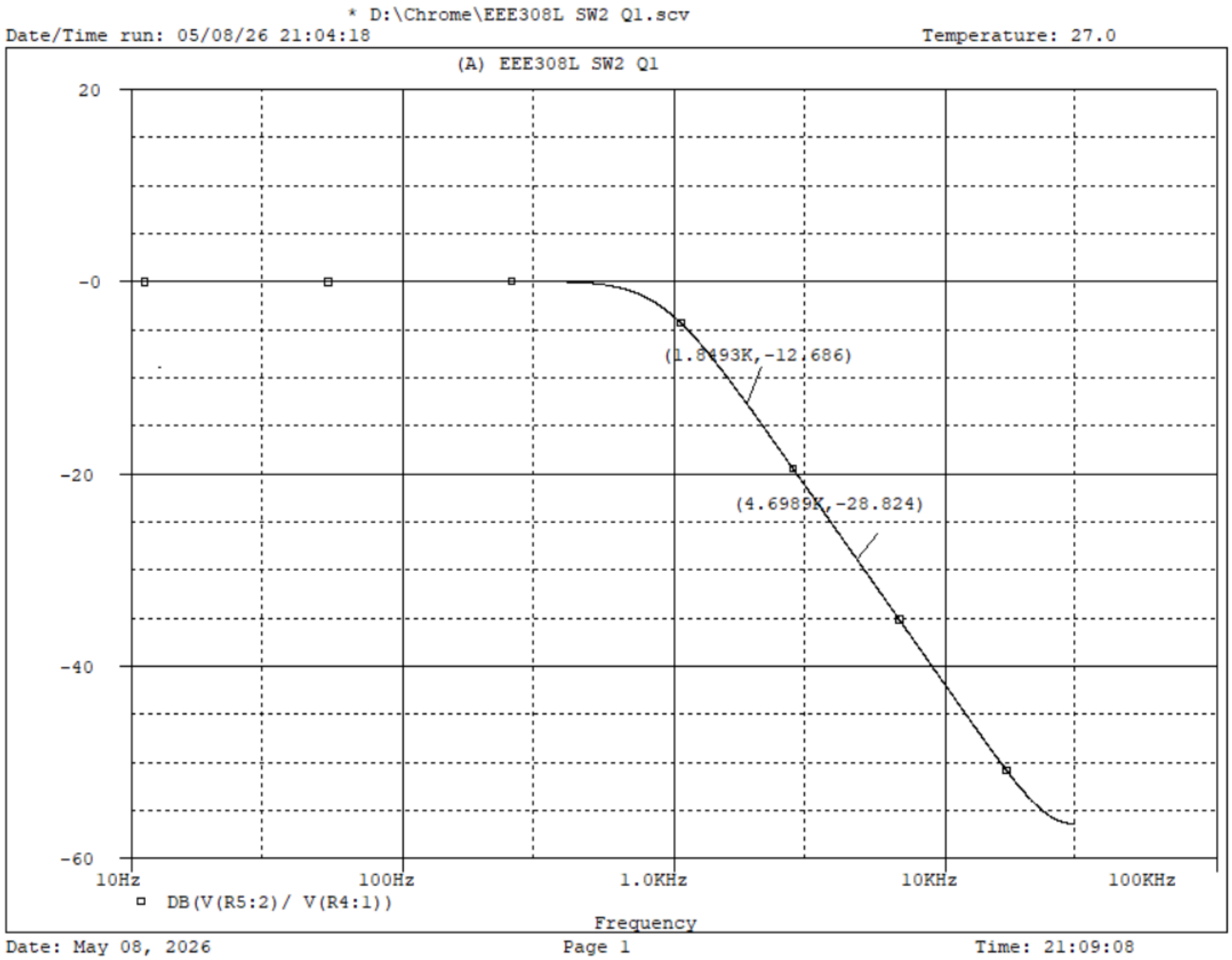


Figure: Graph for Gain (dB) f = 943.641Hz at dB = -3.09

(iii)

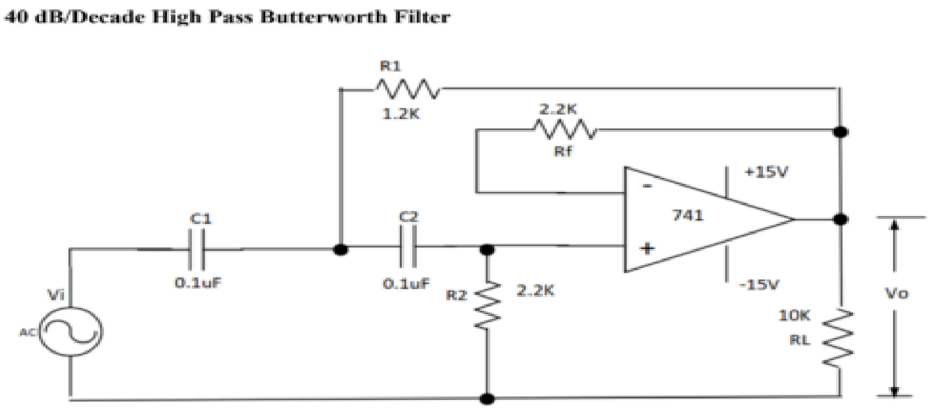


Using 2 different points in the function:

Roll off = $\frac{-28.824-(-12.686)}{log(4.6989)-log(1.8439)}$

= -39.72 ≈ -40dB/Decade (Ans).

Problem-2



Draw the circuit in and give the source 5V amplitude an 0 phase.

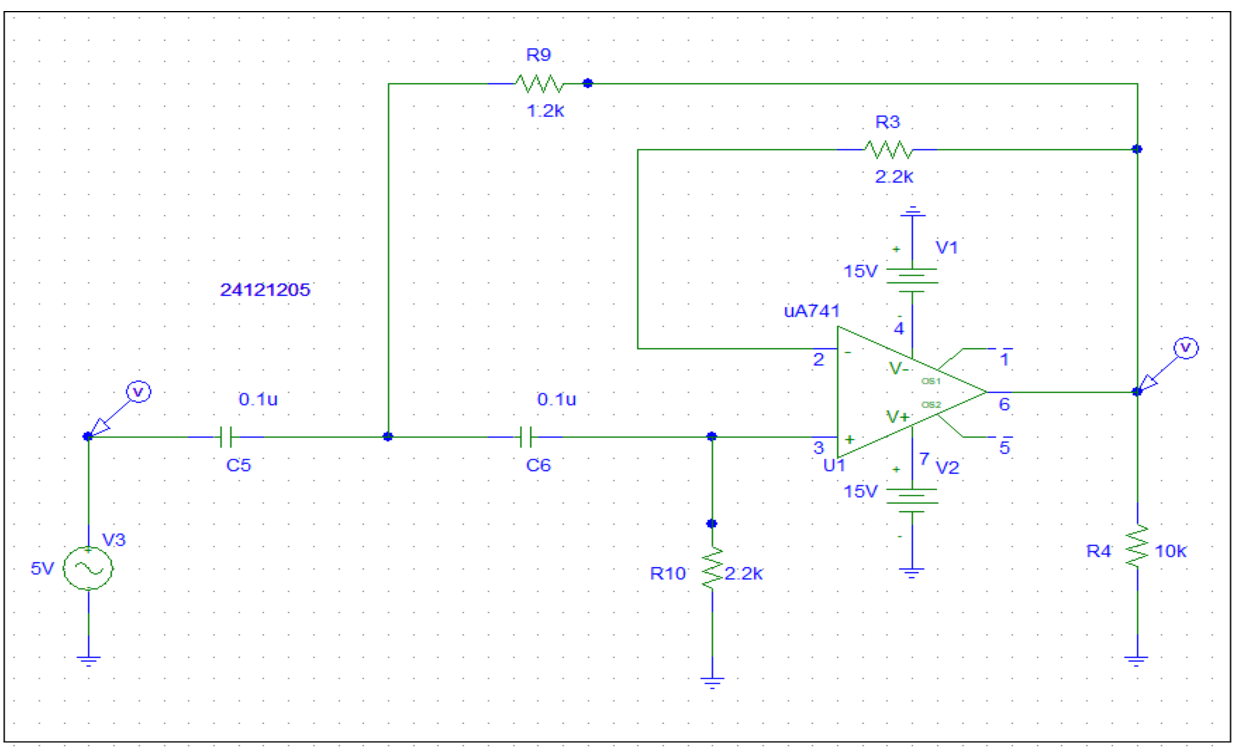
(i) Plot the Gain and Gain(dB) with respect to frequency. USe suitable frequency range.

(ii) From both the Gain and Gain(dB) plot, find the cutoff frequency.

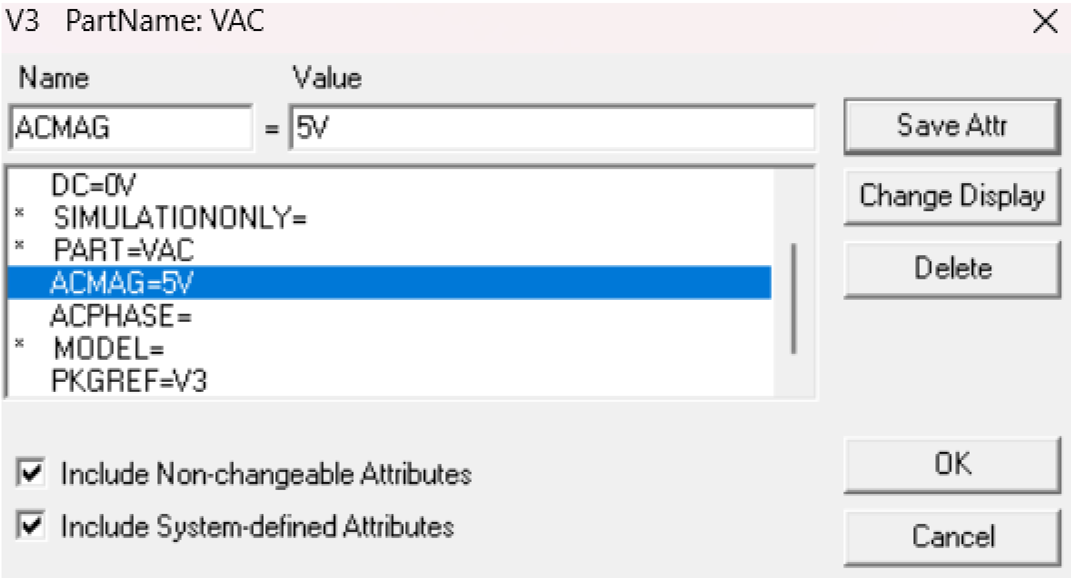
(iii) Show that the roll-off is 40dB/Decade.

Answer:-

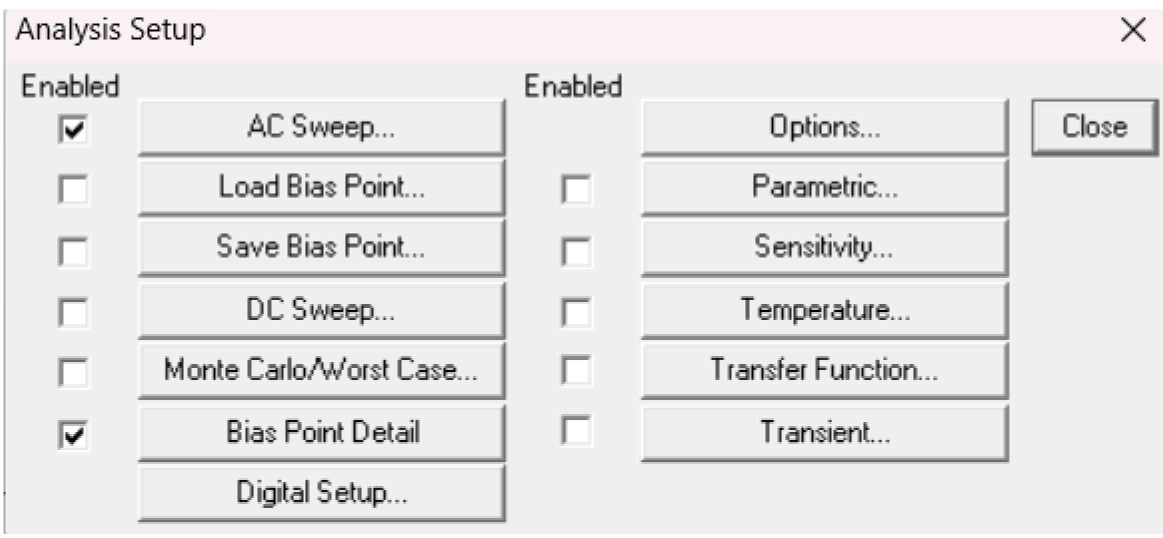
(i) **Circuit Diagram-**



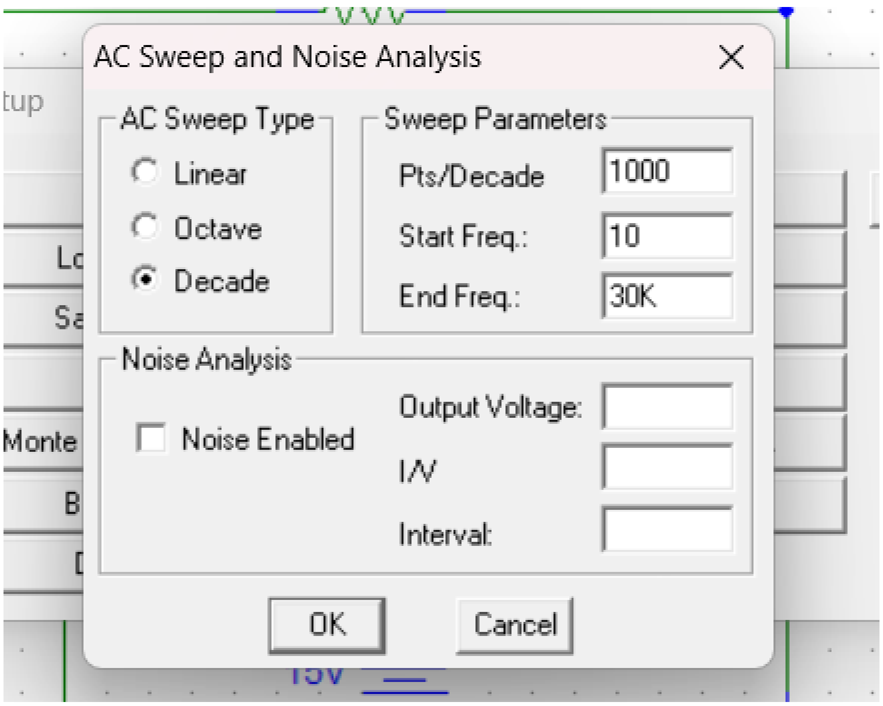
VAC parameter



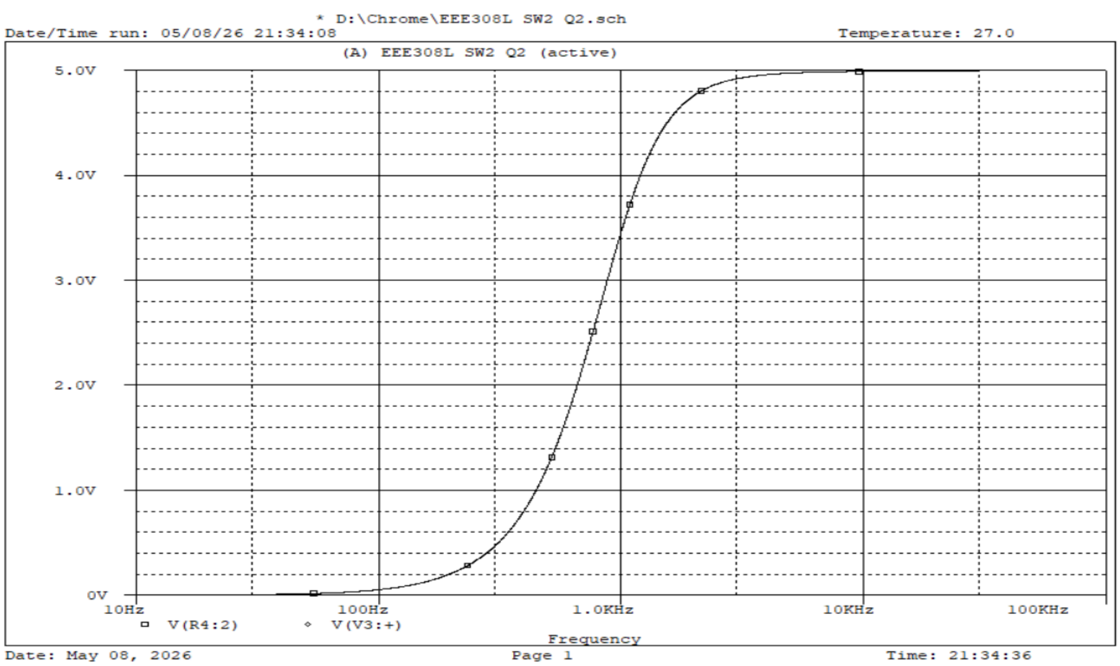
Analysis Setup:



AC Sweep Setup



Graph



(ii)

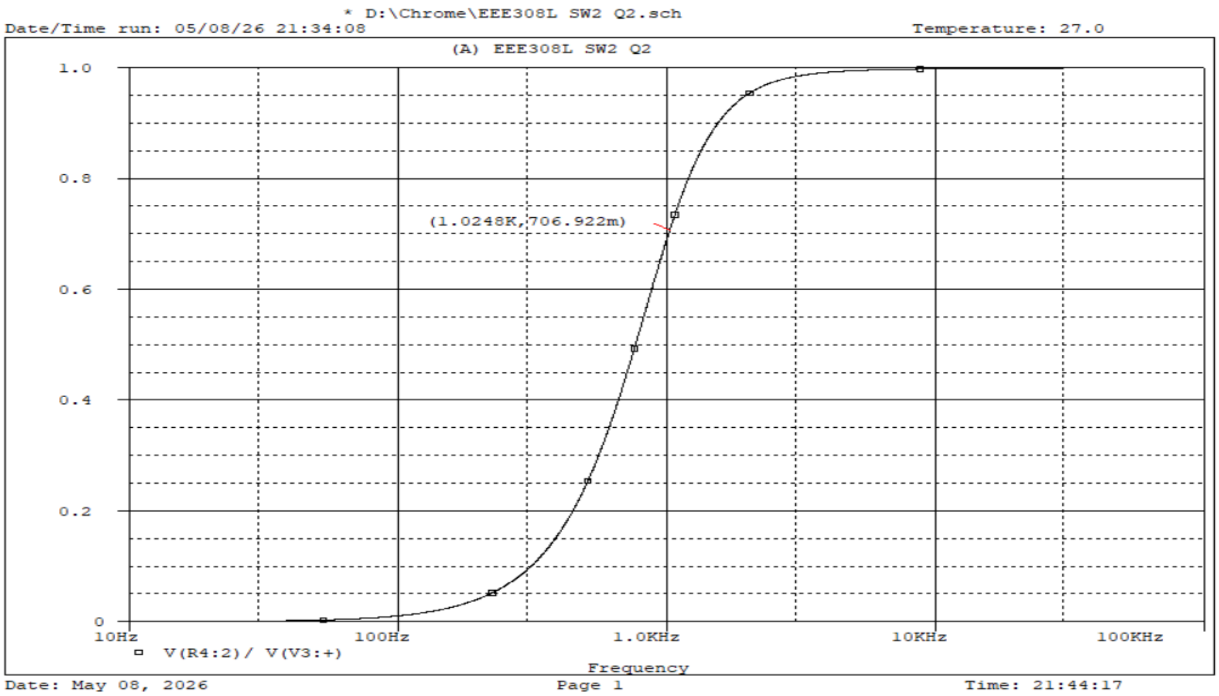


Figure: Graph for Gain (cutoff f= 1.245Hz) at 0.707v Gain

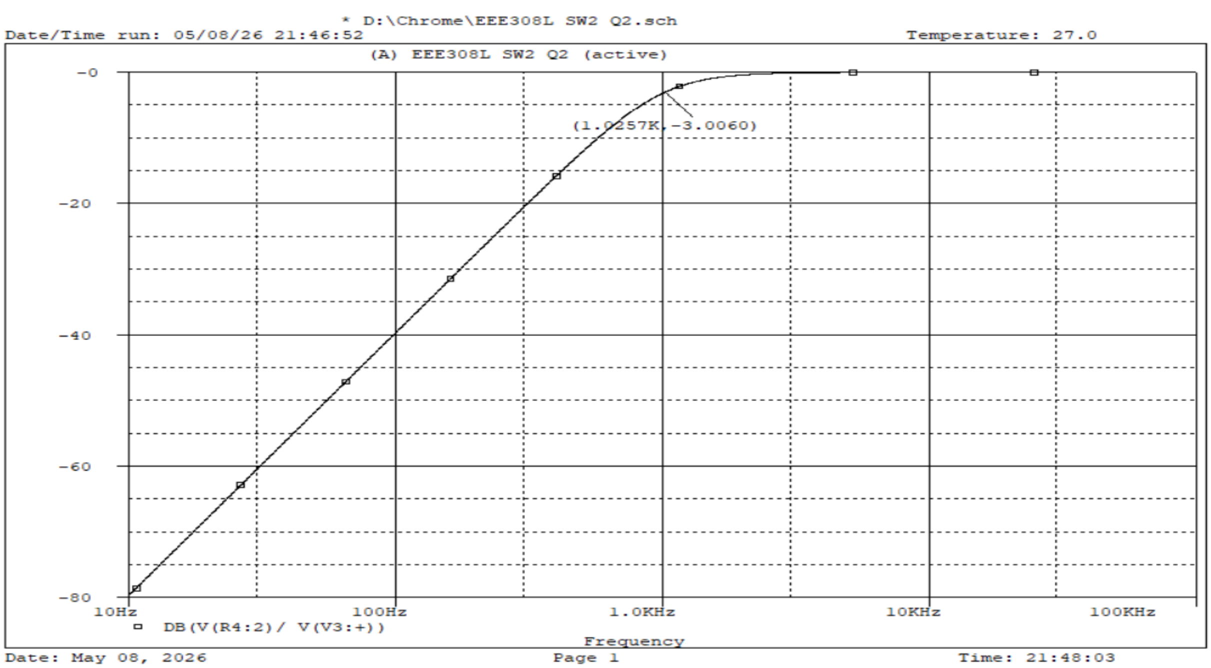
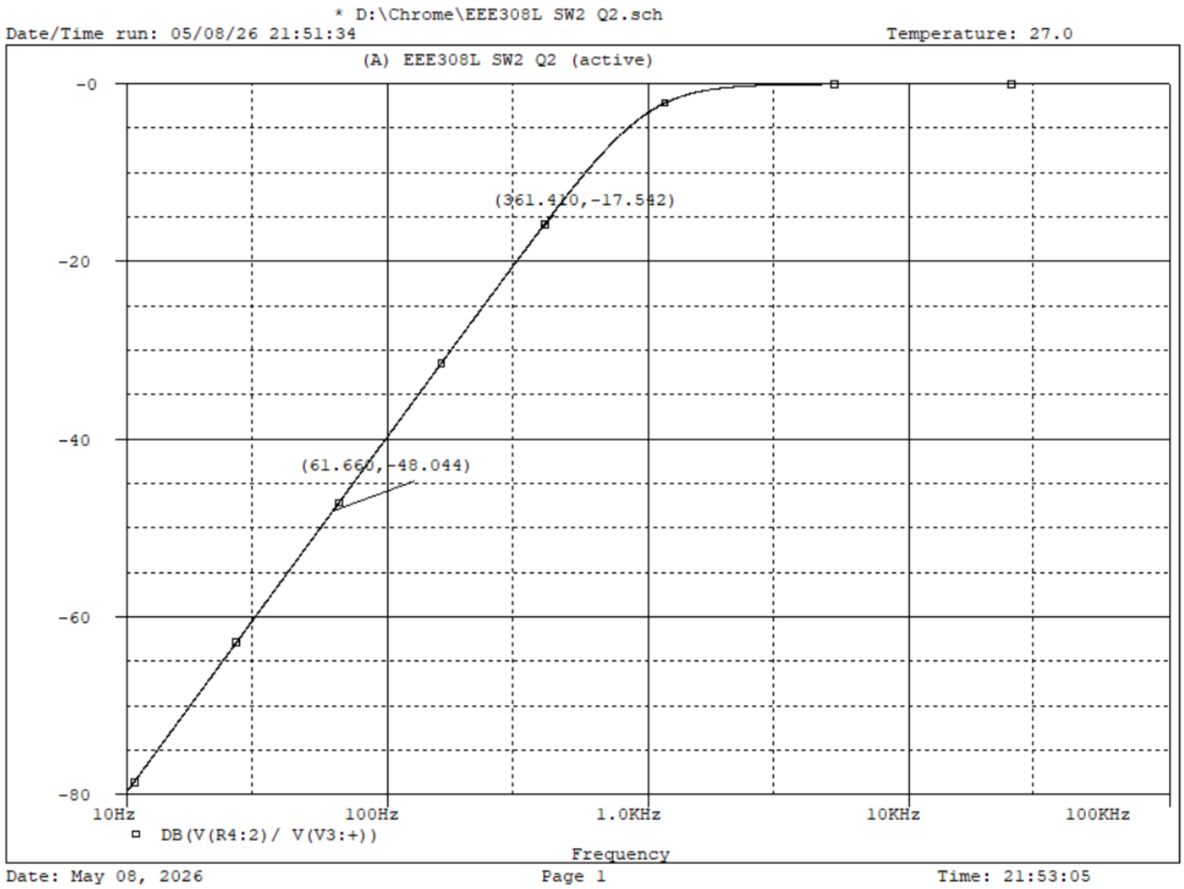


Figure: Graph for Gain (dB) f = 1.0257Hz at dB = -3.006

(iii)

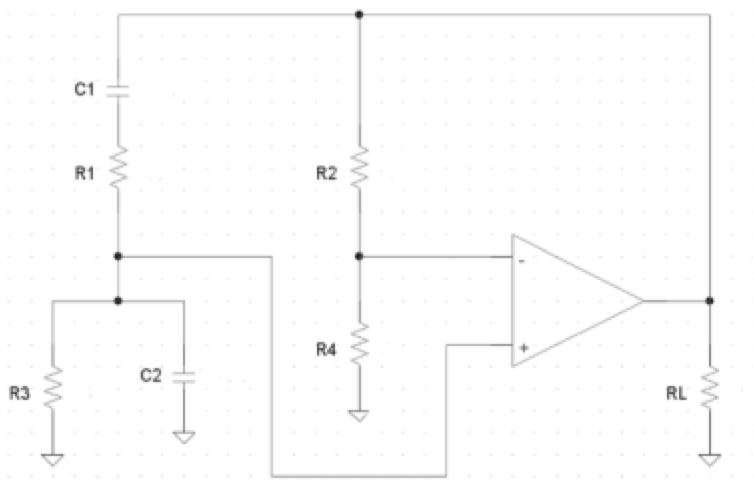


Using 2 different points in the function:

Roll off $= \frac{-48.044-(-17.542)}{log(361.410)-log(61.660)}$

= -39.71 ≈ -40dB/Decade (Ans).

Problem-3



Design the stable sine-wave Wien Bridge oscillator for an audio-testing application. The target frequency is 20x where x = last 2 digits of your ID.

Answer:-

Student Id= 24121205

Here , f= 2005

For Wien Bridge Oscillator, Frequency, f = 1/2πRC

Set, c=0.22uF

R=1/2πfC

= 1/2π*2005*0.22*10^-6

= 3608.137 Ω

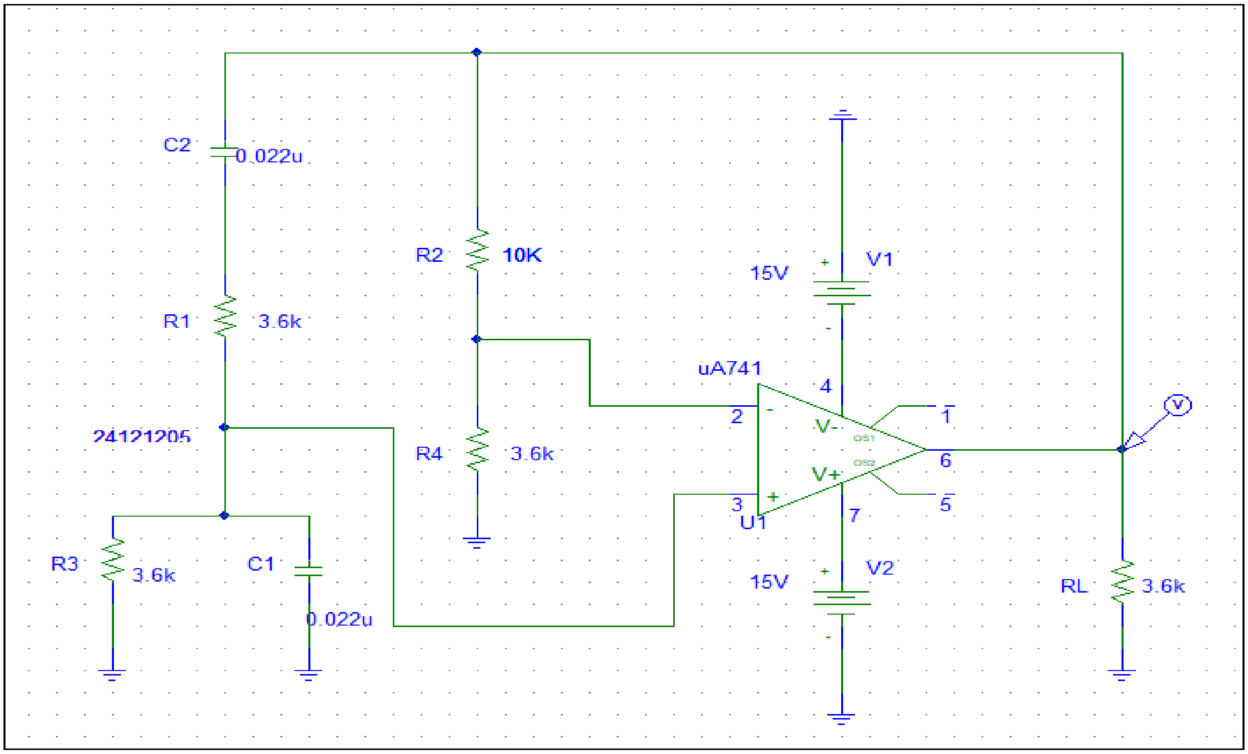
=3.6kΩ

To get a stable oscillation,

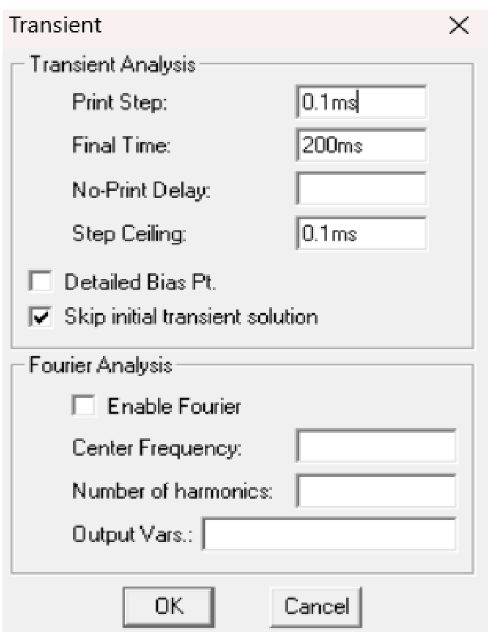
we need a gain over 3, So, we set R2 = 10kΩ

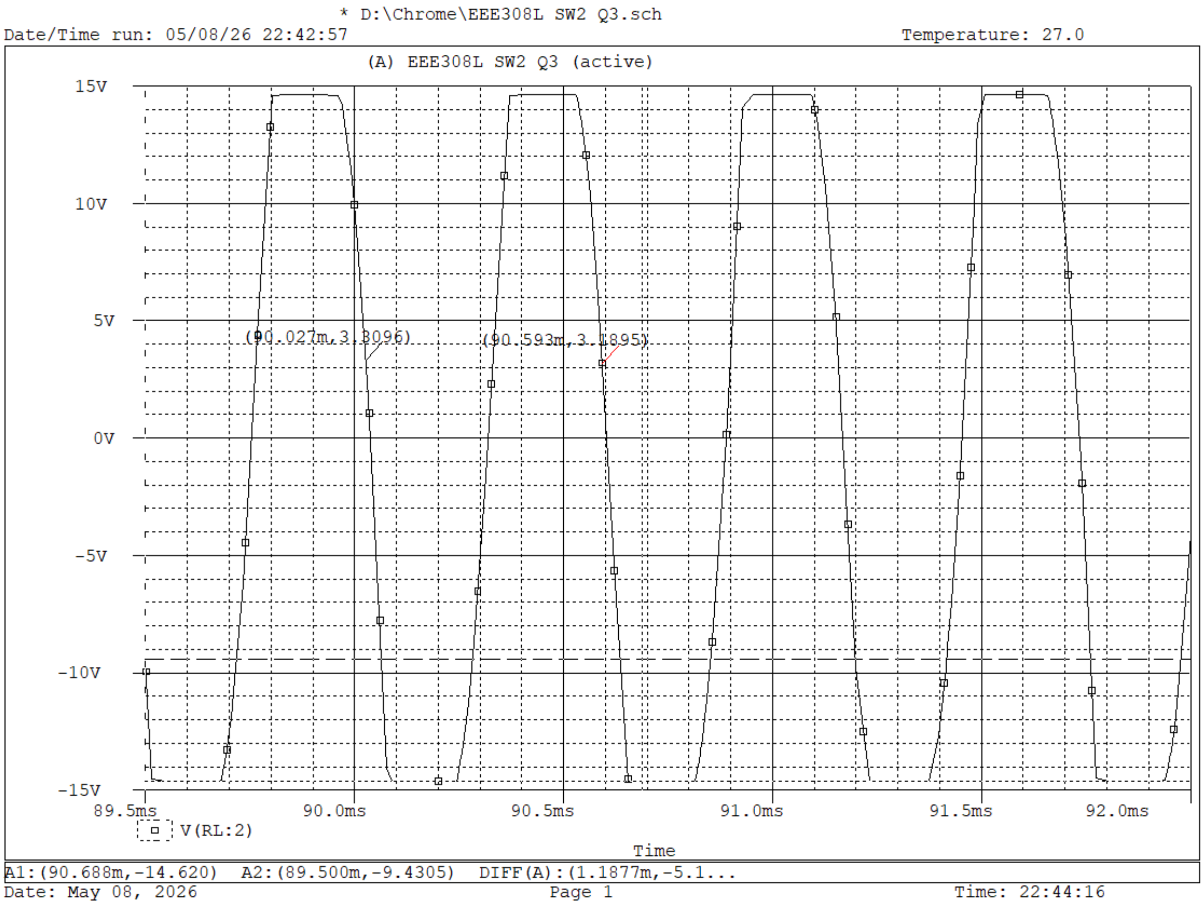
Gain, Av = 1+(R2/R1)=1+(20/3.6)=6.5

Circuit diagram



Transient Setup





Here,

T = (90.593-90.027)ms

So, T = 0.566ms =566 us

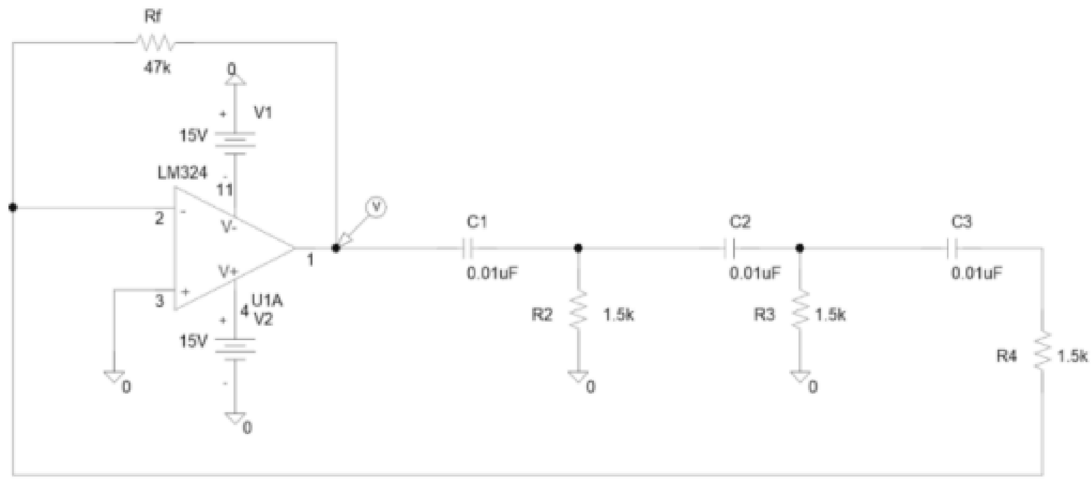
f=1/T=1/566*10^-6

= 1766.8Hz

Error = ((2005-1766.8)/2005)*100%

= 11%

Problem-4



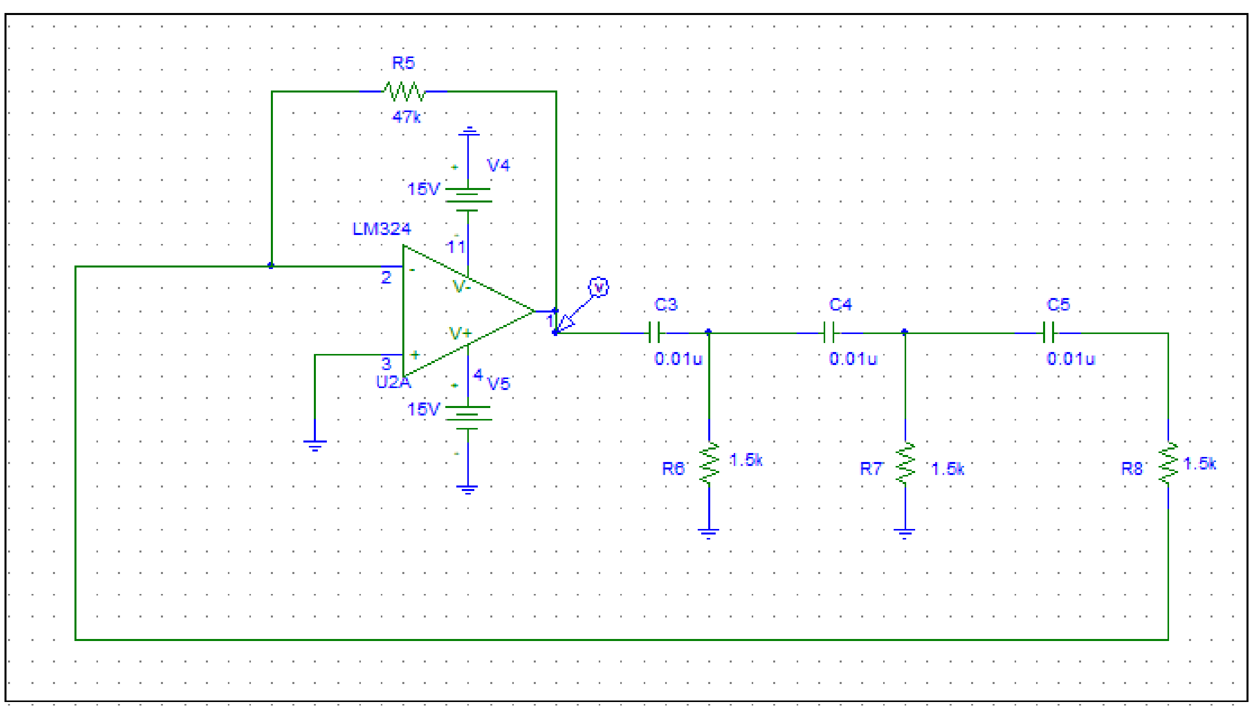
(i) Construct the above circuit.

(ii) Run the simulation in the 'Bias Point Detail' and 'Transient' mode.

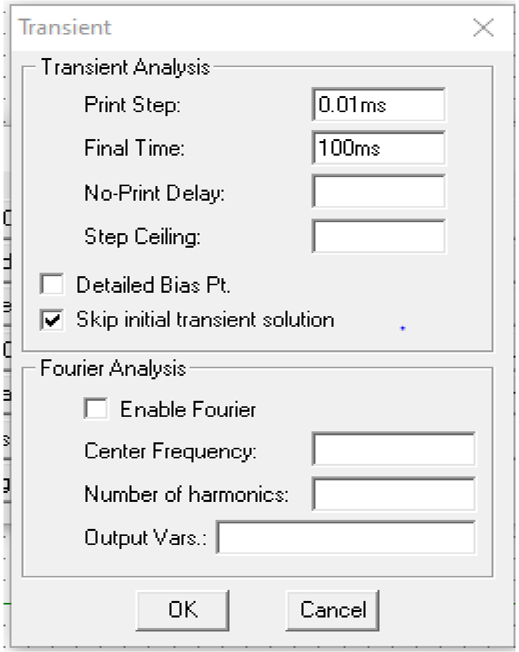
(iii) Observe the wave shapes and determine the frequency.

Answer:-

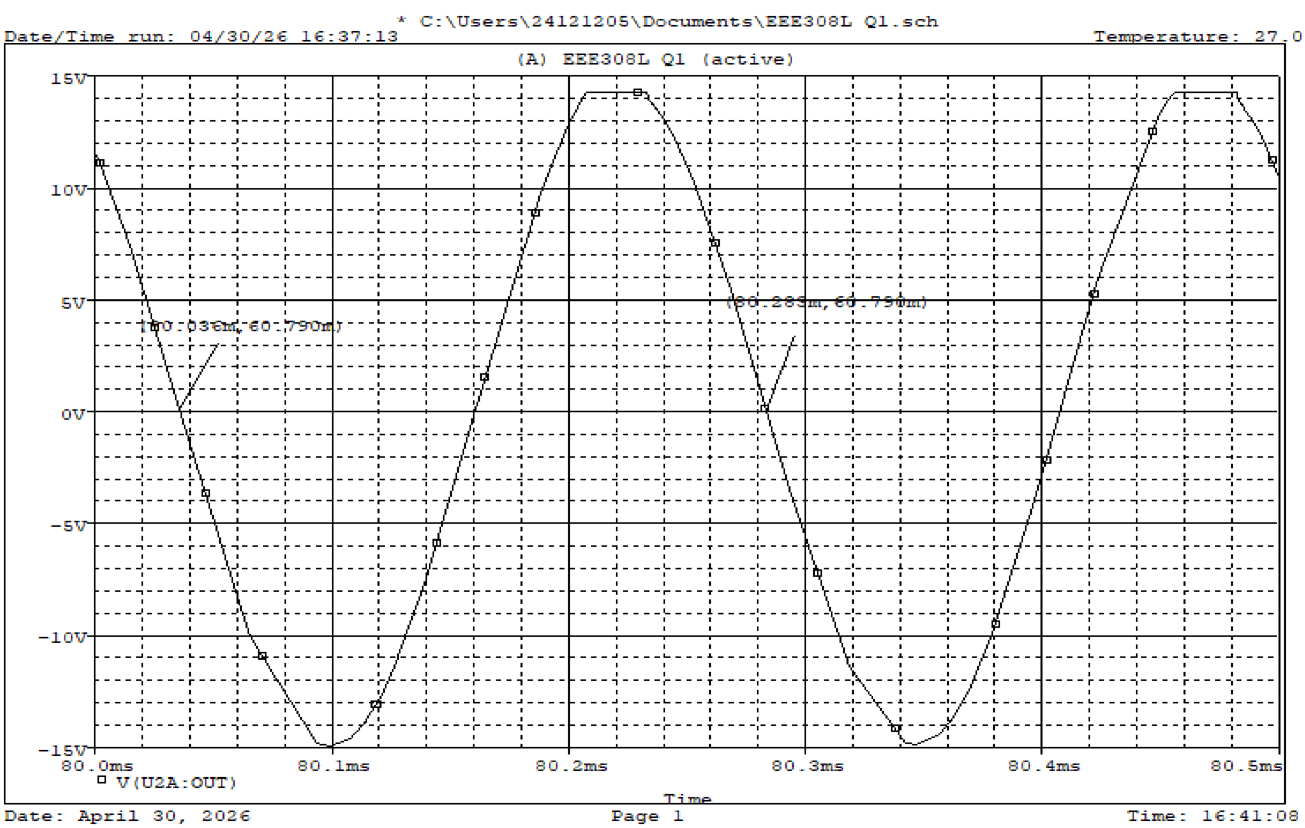
Circuit Diagram:



Transient Setup



V_o Graph



Here,

we find time period between 1 full period,

T = (80.283-80.036)ms

= 0.247ms

= 247us

We know,

frequency, f $=\frac{1}{T}$

$$f = \frac{1}{247*10^{-6}} = 4048.582 Hz (Ans)$$